Shadow complexity lower bounds monotone circuit size for transparent polynomials

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Allowed standard prerequisites (editorial).

- Convex hulls, finite convexly independent sets and arithmetic circuits. Standard background; original shadow, transparency, h.p.-circuit and h.p.-projection definitions retained. Folklore planar Minkowski-sum lemma retained with author outline and original cited source.

Original definitions and computational models (pp.9:1–2,9:11–14)

A polytope is the convex hull of a finite set of points in Euclidean space. Equivalently, it is a compact set that is defined by finitely many linear inequalities. A shadow of a polytope $P \subseteq \mathbb{R}^n$ is a set of the form $L(P)$, where $L : \mathbb{R}^n \to \mathbb{R}^2$ is a linear map. The shadows of P are two-dimensional polygons, and hence typically much simpler than P. The shadow complexity of P is

$$\sigma(P) = \max_L |\operatorname{vert}(L(P))|,$$

where L is a linear map and $\operatorname{vert}(Q)$ is the vertex set of the polytope Q. Let $\mathbb{F}$ be a field. For a list of variables $x = (x_1, \ldots, x_n)$ and $\alpha \in \mathbb{N}^n$, let x^α be the monomial $\prod_{i=1}^n x_i^{\alpha_i}$. A polynomial $f \in \mathbb{F}[x_1, \ldots, x_n]$ is a formal sum of the form $\sum_{\alpha \in \mathbb{N}^n} a_\alpha x^\alpha$ where $\operatorname{supp}(f) := \{\alpha \in \mathbb{N}^n : a_\alpha \neq 0\}$ is finite. The Newton polytope of f is

$$\operatorname{Newt}(f) := \operatorname{conv}(\operatorname{supp}(f)),$$

where $\operatorname{conv}(\cdot)$ denotes the convex hull. A polynomial f is transparent if $|\operatorname{supp}(f)| = \sigma(\operatorname{Newt}(f))$. In other words, there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^2$ which maps $\operatorname{supp}(f)$ to distinct convexly independent points in $\mathbb{R}^2$.

As the standard model of computation of polynomials we take the arithmetic circuit model. It is a (finite) directed acyclic graph whose every node has in-degree zero or two. Nodes of in-degree zero (input nodes) are labelled with a constant or a variable. Nodes of in-degree two are labelled with operations $+$ or $\times$. Every node of the circuit computes a polynomial in $\mathbb{F}$ in the natural way. As the size of the circuit, we take the number of nodes. A circuit is called a formula if its underlying graph is a tree. For more background and motivation, see [44] and references within. Our focus is mainly on monotone computation. A polynomial over $\mathbb{R}$ is monotone if all of its coefficients are non-negative. Similarly, a monotone arithmetic circuit can use only non-negative constants.

In a Laurent polynomial, variables are allowed to have negative integer exponents. The notions of $\mathrm{supp}(f)$ and Newton polytope of f are defined in the obvious manner. A Laurent polynomial over $\mathbb{R}$ is monotone, if all of its coefficients are non-negative. A high powered circuit (h.p.-circuit for short) is an arithmetic circuit in which every input node is labelled by a term $ax_1^{k_1} \cdots x_n^{k_n}$ with $a \in \mathbb{F}$ and $k_1, \ldots, k_n \in \mathbb{Z}$. The size of the circuit is the number of its nodes. In other words, we have given the circuit a power to compute every term ax^α at a unit cost. This is especially important in the case of h.p.-formula. An arithmetic formula of size s can compute a polynomial of degree at most s, whereas there is no such restriction in an h.p.-formula. Furthermore, we have allowed the variables to have negative exponents and hence an h.p.-circuit computes a Laurent polynomial instead of a polynomial. But this is only a cosmetic detail.

A high power projection (h.p.-projection for short) is a homomorphism

$$\pi : \mathbb{F}[x_1, \ldots, x_n] \to \mathbb{F}[y_1, \ldots, y_m, y_1^{-1}, \ldots, y_m^{-1}]$$

such that $\pi(x_i) = a_i y^{\alpha_i}$ for every x_i, where $a_i \in \mathbb{F}$ and $\alpha_i \in \mathbb{Z}^m$, and for every $f \in \mathbb{F}[x_1, \ldots, x_n]$,

$$\pi(f(x_1, \ldots, x_n)) = f(\pi(x_1), \ldots, \pi(x_n)).$$

The constants a_i are called the coefficients of π and α_i its exponents. If $\mathbb{F} = \mathbb{R}$ and π has non-negative coefficients, then π is called monotone. An h.p.-projection π induces a linear map $L_\pi : \mathbb{R}^n \to \mathbb{R}^m$ by setting $L_\pi(e_i) = \alpha_i$ and extending it linearly to $\mathbb{R}^n$. It follows that $\mathrm{supp}(\pi(f)) \subseteq L_\pi(\mathrm{supp}(f))$. The inclusion may be strict, as some monomials of f can cancel out in the projection. If no cancellations occur, we indeed have $\mathrm{Newt}(\pi(f)) = L_\pi(\mathrm{Newt}(f))$. This is satisfied, e.g., if f is monotone and the coefficients of π are positive, or if the coefficients are algebraically independent.

Theorem 2 (p.9:2). *If f is transparent then every monotone circuit computing f has size at least $\Omega(\sigma(\mathrm{Newt}(f)))$.*

Original bearing geometric statements (pp.9:6,9:12,9:14–15)

Given $A, B \subseteq \mathbb{R}^n$, their Minkowski sum is defined as

$$A + B := \{a + b : a \in A, b \in B\}.$$

If P and Q are polytopes then $P+Q$ is also a polytope. In two-dimensions, Minkowski sum has nice properties. Let P be a polytope in $\mathbb{R}^2$ with vertices $v_1, \ldots, v_k$ where $k > 1$. We can assume they are ordered so that P lies in the left closed half plane determined by the line going from v_i to v_{i+1} for $i < k$, and similarly for v_k and v_1. Let $E(P)$ be the collection of unit vectors in the direction of these k edges. That is, vectors of the form $(v_{i+1} - v_i)/\|v_{i+1} - v_i\|$ for $i < k$, and $(v_1 - v_k)/\|v_1 - v_k\|$. If $|\operatorname{vert}(P)| \leq 1$ then $E(P) := \emptyset$.

Lemma 8. *Let $P_1, \ldots, P_r$ be non-empty polytopes in $\mathbb{R}^2$. Then $E(P_1 + \cdots + P_r) = \bigcup_{i=1}^r E(P_i)$. Consequently, $|\operatorname{vert}(P_1 + \cdots + P_r)| \leq \sum_{i=1}^r |\operatorname{vert}(P_i)|$. The latter holds for empty P_i's as well.*

The lemma is folklore. It can be inferred from Chapter 13.3 in [11], and we give only an outline of proof.

Proof sketch of Lemma 8. Given a non-empty polytope P and $w \in \mathbb{R}^2$, let $P^w := \{x \in P : \langle x, w\rangle = \max_{z \in P}\langle z, w\rangle\}$ be the set of extreme of points of P in the direction w. It is either a vertex or an edge of P. For a pair of polytopes we have $(P_1+P_2)^w = P_1^w + P_2^w$. Every edge of P_1 yields an edge of P_1+P_2 with the same direction. Conversely, every edge of $P_1 + P_2$ comes from one of P_1 or P_2. □

Lemma 28. *Let f be a polynomial over an infinite field $\mathbb{F}$. Assume that $\sigma(\operatorname{Newt}(f)) = k$. Then there exists a bivariate Laurent polynomial $g \in \mathbb{F}(y_1, y_2, y_1^{-1}, y_2^{-1})$ which is an h.p.-projection of f so that $\operatorname{Newt}(g)$ has k vertices. Moreover, if f is a homogeneous polynomial then g is a polynomial. If $\mathbb{F} = \mathbb{R}$, then the coefficients in the projection can be assumed positive.*

Proof. Let $L(z) = Az$ with $A \in \mathbb{R}^{2\times n}$ be a linear map so that

$$|\operatorname{vert}(L(\operatorname{Newt}(f)))| = k.$$

We can assume that the entries of A are rational, because a small perturbation of A cannot decrease $|\operatorname{vert}(L(\operatorname{Newt}(f)))|$. Now, we can assume that the entries of A are integers, because we can multiply A by a suitable integer. Moreover, when f is homogeneous of degree d, increasing all entries of A by λ corresponds to shifting $L(\operatorname{Newt}(f))$ by $(\lambda d, \lambda d)$, which does not change the number of vertices. Hence, in this case, A can be taken with non-negative integer entries. Let us now consider a h.p.-projection π with $\pi(x_i) = a_i y^{L(e_i)}$. It follows that $\operatorname{supp}(\pi(f)) \subseteq L(\operatorname{supp}(f))$. Now, we claim that we can choose the coefficients a_i so that equality holds. This can be seen as follows. Given $\alpha \in \operatorname{supp}(f)$, the coefficient of $y^{L(\alpha)}$ in $\pi(f)$ is a non-zero polynomial h_α in the coefficients of π. Hence, if $\mathbb{F}$ is infinite, there exist non-zero coefficients for which h_α is non zero for every $\alpha \in \operatorname{supp}(f)$. If $\mathbb{F} = \mathbb{R}$, they can be taken positive. □

Lemma 37. *Let $A, B \subseteq \mathbb{R}^2$ be non-empty sets such that $A + B$ is a convexly independent set with $|A| \geq |B|$. Then either $|A| \leq 2$ or $|B| \leq 1$.*

Proof. For the sake of contradiction, assume that $A+B$ is convexly independent, $|A| \geq 3$ and $|B| \geq 2$. By Lemma 8, the convex hull of $A+B$ has at most $|A|+|B|$ vertices. By the size assumption, there exist $a_1 \neq a_2 \in A$ and $b_1 \neq b_2 \in B$ with $a_1 + b_1 = a_2 + b_2$. The point $a_1 + b_1$ is the average of $a_1 + b_2$ and $a_2 + b_1$ and it is distinct from them, a contradiction. □

Entire original circuit argument (Section 5.5, pp.9:15–16)

Theorem 38. *Let f be a monotone bivariate Laurent polynomial such that* $\operatorname{supp}(f)$ *is convexly independent and* $|\operatorname{supp}(f)| = k$*. Then f requires monotone h.p.-circuit with $k/4$ gates.*

Theorem 38 implies Theorem 2 via Lemma 28.

Proof. The lower bound is proved using the following "progress" measure. Given $A \subseteq \mathbb{R}^2$ and $\epsilon \in \{0,1\}$, let $A^\epsilon := A$ if $\epsilon = 1$ and $A^\epsilon := \emptyset$ if $\epsilon = 0$. Given $v \in \mathbb{R}^2$, let $v + A := \{v\} + A$. Let $\mathcal{A}$ be a finite set of finite subsets of $\mathbb{R}^2$. For functions $\epsilon : \mathcal{A} \to \{0,1\}$ and $v : \mathcal{A} \to \mathbb{R}^2$, let

$$\mathcal{A}_{\epsilon,v} = \bigcup_{A \in \mathcal{A}} (v(A) + A)^{\epsilon(A)}.$$

Let

$$\mu(\mathcal{A}) = \max_{\epsilon,v} \{|\mathcal{A}_{\epsilon,v}| : \mathcal{A}_{\epsilon,v} \text{ is convexly independent}\}.$$

Claim. Let $\mathcal{A}' = \mathcal{A} \cup \{B\}$ and $A_1, A_2 \in \mathcal{A}$. Then

$$\mu(\mathcal{A}') \le \mu(\mathcal{A}) + |B|, \tag{3}$$
$$\mu(\mathcal{A}') \le \mu(\mathcal{A}) + 2, \qquad \text{if } B = u + A_1 \text{ for some } u \in \mathbb{R}^2, \tag{4}$$
$$\mu(\mathcal{A}') \le \mu(\mathcal{A}) + 4, \qquad \text{if } B = A_1 \cup A_2, \tag{5}$$
$$\mu(\mathcal{A}') \le \mu(\mathcal{A}) + 4, \qquad \text{if } B = A_1 + A_2. \tag{6}$$

Proof of Claim. Inequality (3) is straightforward. To prove (4), suppose that ϵ, v are such that $\mathcal{A}'_{\epsilon,v}$ is convexly independent. Suppose $\epsilon(A_1) = \epsilon(B) = 1$ and $v(A_1) + A_1 \neq v(B) + B$; otherwise we have $|\mathcal{A}'_{\epsilon,v}| \le \mu(\mathcal{A})$. Then $(v(A_1) + A_1) \cup (v(B) + B) = \{v(A_1), v(B) + u\} + A_1$ is convexly independent. Since $|\{v(A_1), v(B) + u\}| = 2$, by Lemma 37, A_1 has size at most 2. This means $\mu(\mathcal{A}') \le \mu(\mathcal{A}) + 2$ by (3). For (5), observe that $\mu(\mathcal{A}') \le \mu(\mathcal{A} \cup \{u_1 + A_1, u_2 + A_2\})$ whenever $u_1, u_2 \neq 0$ are distinct and apply (4) twice. Finally, we prove (6). If $B = A_1 + A_2$ is not convexly dependent, it contributes nothing to μ. Assume that B is convexly independent and $|A_1| \ge |A_2| > 0$. By Lemma 37, either $|A_1 + A_2| \le 4$ or $|A_2| = 1$. In the former case, $\mu(\mathcal{A}') \le \mu(\mathcal{A}) + 4$ by (3). In the latter, $A_2 = \{u\}$ for some u and $\mathcal{A}' = \mathcal{A} \cup \{u + A_1\}$ and we can apply (4). □

Let us call a h.p.-circuit transparent, if every gate in the circuit computes a polynomial with convexly independent support. Given a circuit Ψ and a node u, let $\operatorname{supp}(u)$ be the support of the Laurent polynomial computed by u. Let $\mathcal{A}_\Psi$ be the set $\{\operatorname{supp}(u) : u \in \Psi\}$. Using the Claim, we can show that whenever a transparent and monotone Ψ has s gates then $\mu(\mathcal{A}_\Psi) \le 4s$. The proof is by induction. The induction base $s = 1$ trivially holds. It remains to verify the induction step. Let u be an output gate of Ψ. If u is also an input gate, apply (3). If $u = u_1 + u_2$ then $\operatorname{supp}(u) = \operatorname{supp}(u_1) \cup \operatorname{supp}(u_2)$ and (5) completes the proof. If $u = u_1 \times u_2$ then $\operatorname{supp}(u) = \operatorname{supp}(u_1) + \operatorname{supp}(u_2)$ and (6) completes the proof. Finally, consider a monotone circuit Ψ for f of minimal size s. No gate in the circuit computes the zero polynomial (unless f itself the zero polynomial).

The circuit is transparent because a monotone computation does not cancel monomials unless multiplying by zero, and because $+, \times$ can not "undo" convex dependence. This means that $\mu(\mathcal{A}_\Psi) \leq 4s$. On the other hand, since $\mathrm{supp}(f)$ consists of k convexly independent points, we have $\mu(\mathcal{A}_\Psi) \geq |\mathrm{supp}(f)| = k$. $\square$

References

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